Muhammet Celal Gokalp

Systems Engineer | Autonomy Integration & Verification

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PROFESSIONAL SUMMARY

Systems engineer with nearly 8 years on safety-related automotive systems, from driver assistance features at Jaguar Land Rover to an L4 automated driving stack at FiveAI / Bosch. Experienced in turning stakeholder needs into verifiable requirements, defining system states and acceptance criteria, and building scenario-based verification in simulation, HiL/ViL and real vehicles, with traceability back to requirements. Used to working across teams to find the root cause of integration issues. Strong at reading Python and comfortable reading C++. No ROS 2 experience yet, but I learn new stacks quickly and autonomous vehicles share many of the same problems as mobile robots.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – Present

FiveAI / Bosch

Cambridge, United Kingdom

- **Scenario-based V&V:** Design test scenarios for an L4 automated driving stack in OpenSCENARIO 2, from functional to logical to concrete. Built 120+ parameterised scenarios that generate thousands of test cases, covering Euro NCAP junction tests and UN R79 automatic lane change on the ADA project.
- **Acceptance criteria & performance:** Defined and checked pass/fail and KPI criteria, such as comfort limits for acceleration and deceleration across 1,000+ hours of test data, and fed results back to the development teams.
- **Development environment:** Worked in the ADA project monorepo, where scenarios were stored and run. Ran simulations locally through the project's dev container and through the test pipeline (dtflow), reading Python and C++ code to understand stack behaviour and failures.
- **Cross-team integration:** On the ADA project, a partnership with Volkswagen Group where FiveAI supplied the simulation tooling, trained partner engineers to use our toolchain. This grew into the Simulation EcoSystem solutions team, where I was a founding member. Triaged integration issues across tools, maps (SDB/XODR) and simulation, working with several engineering teams to keep enterprise customers running.
- **Tooling:** Built internal tools (a GUI for JSON test parameters, a local uploader) and data pipelines that send simulation results to front-end dashboards, using AI-assisted coding and raising pull requests to the main code base.

ADAS Systems Engineer (Feature Owner)

11/2018 – 10/2021

AVL Powertrain UK

Coventry, United Kingdom

- **Requirements & architecture:** Based at Jaguar Land Rover as feature owner for Rear Cross Traffic Alert, Blind Spot Monitor and Rear Pre-Crash. Wrote use cases, system state diagrams and performance requirements, and broke high-level requirements down into low-level ones in DOORS and MKS Integrity (ASPICE V-model).
- **Verification & traceability:** Wrote test cases for each system state, traced to requirements, and ran verification on HiL and ViL rigs and on the proving ground.
- **System-level investigation:** Used Vector CANoe and CANape to analyse vehicle network data. For example, when an alert triggered late at boundary conditions, I re-ran the tests, found the cause in the logs, and agreed a sensor field-of-view change with the supplier while keeping false alerts low.
- **Field diagnostics:** At the LEVC (London Taxi) factory, inspected returned faulty e-machines: multimeter checks, downloading data from proprietary loggers and sending it to developers and suppliers for root cause analysis.
- **Safety:** Wrote ISO 26262 functional safety requirements for a lane change feature. Trained in ISO 26262 and SOTIF (ISO 21448).

EDUCATION

MSc in Sustainable Automotive Engineering

10/2017 – 09/2018

Warwick Manufacturing Group, Coventry, UK

- **Grade:** 2:1 (Merit). Dissertation: experimental testing to find the variables of a thermal model of an electric motor stator (MATLAB, Simulink, MotorCAD).

BSc in Mechanical Engineering

08/2011 – 02/2017

Ege University, Izmir, Türkiye

- Dissertation: designed, built and controlled a 6-DOF Stewart motion platform, a small robotic system (SolidWorks, MATLAB, Arduino).

SKILLS

Systems Engineering	Requirements decomposition, use cases, system state diagrams, acceptance criteria, traceability, V-model / ASPICE, DOORS, MKS Integrity
Verification	Scenario-based testing, simulation, SIL / HiL / ViL, proving ground testing, KPI analysis, regression testing
Safety	ISO 26262 (requirements written, trained), SOTIF / ISO 21448 (trained)
Software & Tools	Python (strong reading, scripting), C++ (reading), OpenSCENARIO 2, Git, monorepo and dev containers, MATLAB / Simulink, Vector CANoe / CANape, Jira
Languages	English (C2), Turkish (Native)